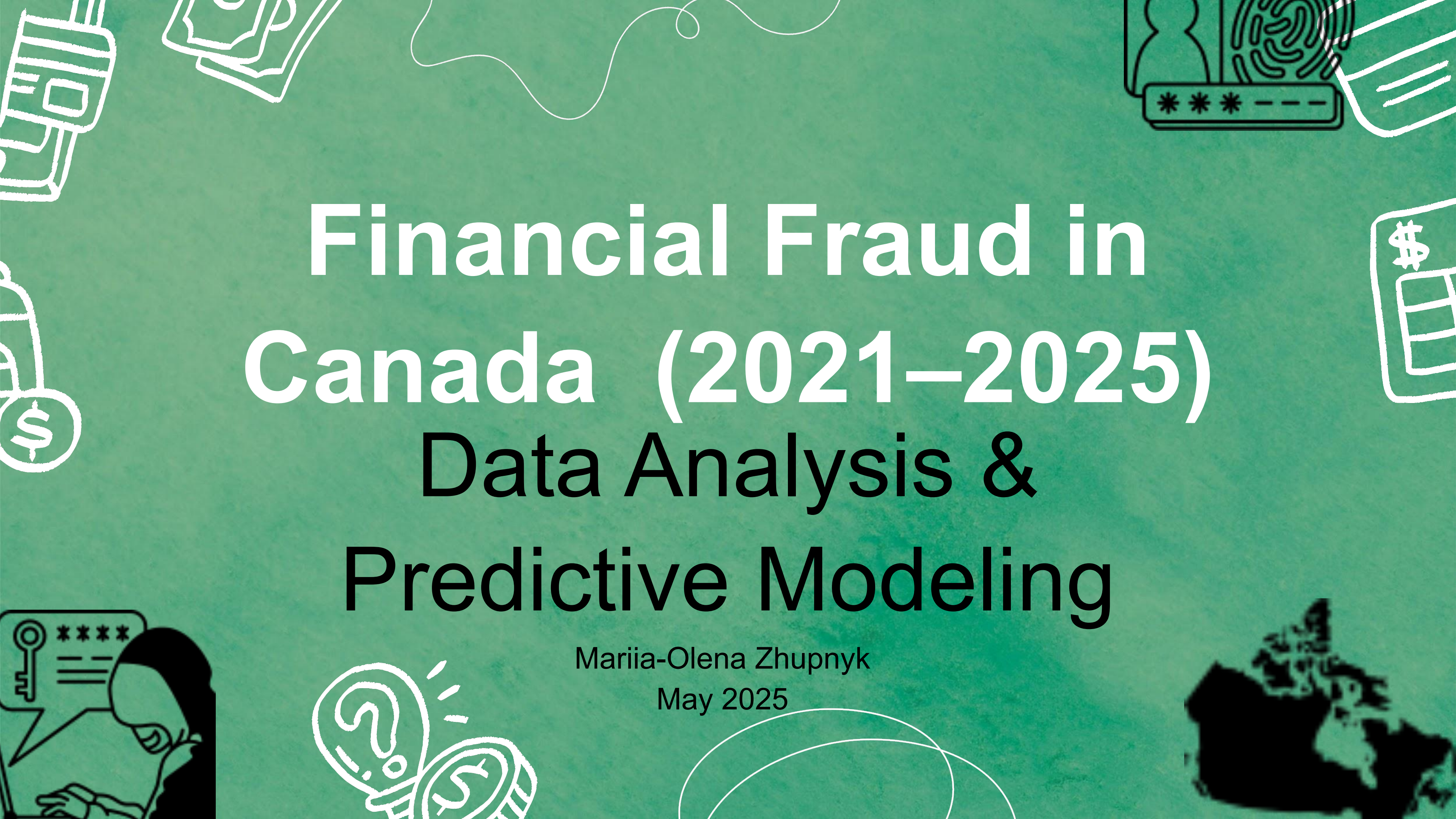


The background is a textured teal color. It is decorated with several white line-art icons: a stack of money in the top left, a fingerprint scanner and a card with three asterisks in the top right, a dollar sign in a circle on the left, a document with a dollar sign on the right, a person at a computer with a key icon and five asterisks in the bottom left, a lightbulb with a question mark and a dollar sign in the bottom center, and a map of Canada in the bottom right. There are also some faint white circles at the bottom.

Financial Fraud in Canada (2021–2025) Data Analysis & Predictive Modeling

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Data Source & Tools

- Source: [Government of Canada - Open Data](#)
- Period Covered: Jan 2021 – Mar 2025
- Scope: Fraud type, reported losses, dates, location, demographics
- Tools: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Plotly)
- Jupyter Notebook, GoogleColab

```
1 # Preview data
2 print("Initial shape:", df.shape)
3 df.head()
```

Initial shape: (328649, 21)

	Número d'identification / Number ID	Date Received / Date reçue	Complaint Received Type	Type de plainte reçue	Country	Pays	Province/State
0	1	2021-01-02	CAFC Website	CAFC site web	Canada	Canada	Saskatchewan
1	2	2021-01-02	CAFC Website	CAFC site web	Not Specified	Non spécifié	Not Specified
2	3	2021-01-02	CAFC Website	CAFC site web	Canada	Canada	Quebec
3	4	2021-01-02	CAFC Website	CAFC site web	Canada	Canada	Saskatchewan
4	5	2021-01-02	CAFC Website	CAFC site web	Canada	Canada	Saskatchewan

5 rows × 21 columns

Project Objective

- Understand patterns in fraud complaints
- Identify high-risk groups and fraud methods
- Analyze financial loss trends
- Explore patterns in fraud reports and build predictive models to identify whether a reported incident involved a victim who lost money.

Why It Matters:

Fraud affects thousands of Canadians annually—this analysis helps reveal where and when it's happening most.



Data Cleaning & Preparation

- Remove Duplicate Columns (Keep English only)
- Rename columns for clarity,
- Removing extra characters and whitespace,
- Converting Dollar_Loss to numeric,
- Converting Date_Received to the correct format
- Standardizing age ranges and other categorical values,
- Handling missing or unspecified entries properly.

```
2 df = df.replace(['Not Available', 'non disponible', 'Not Specified', 'Autre/inconn'])

1 # Check missing values
2 missing_summary = df.isna().sum()
3 missing_summary
```

	0
ID	0
Date_Received	0
Complaint_Source	0
Country	74367
Province	77767
Fraud_Category	0
Solicitation Method	17387
Gender	84113
Language of Correspondence	84555
Victim_Age_Range	98111
Complaint Type	0
Victim_Count	0
Dollar_Loss	0

dtype: int64

```
[ ] 1 missing_summary = df.isna().sum().sort_values(ascending=False)
     2 print("\nMissing values summary:\n", missing_summary[missing_summary > 0])
```

```
Missing values summary:
Victim_Age_Range      98111
Language of Correspondence 84555
Gender                84113
Province              77767
Country               74367
Solicitation Method   17387
dtype: int64
```

Exploratory Data Analysis

How is Financial Fraud Changing Over Time?

The line chart shows a steady decline in reported financial fraud cases from 2021 to early 2025.

- Reports peaked in early 2021 at over 11,000 per month.
- By 2024, the number of monthly reports stabilized at a much lower level, around 4,000 to 5,000 cases.

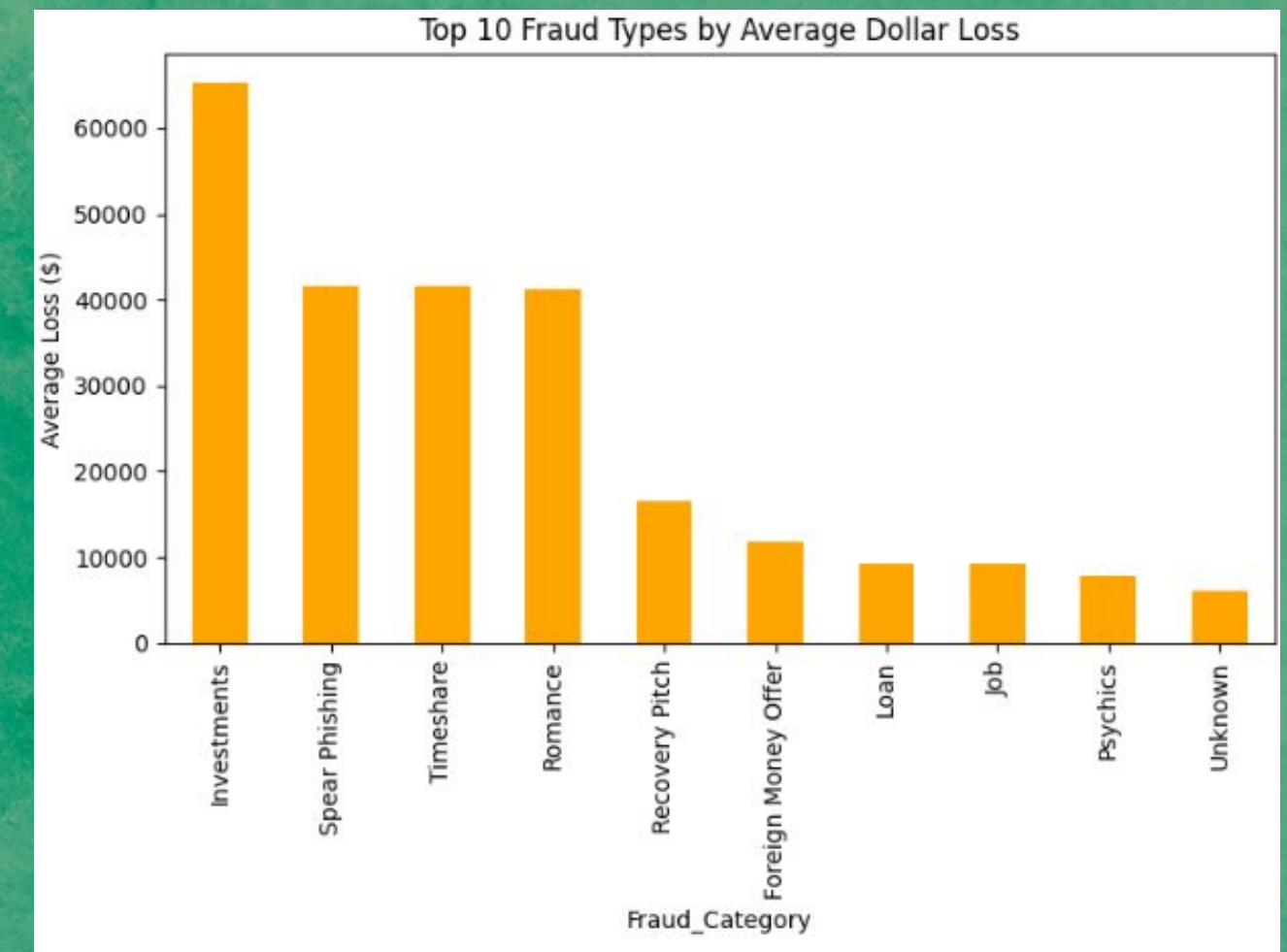
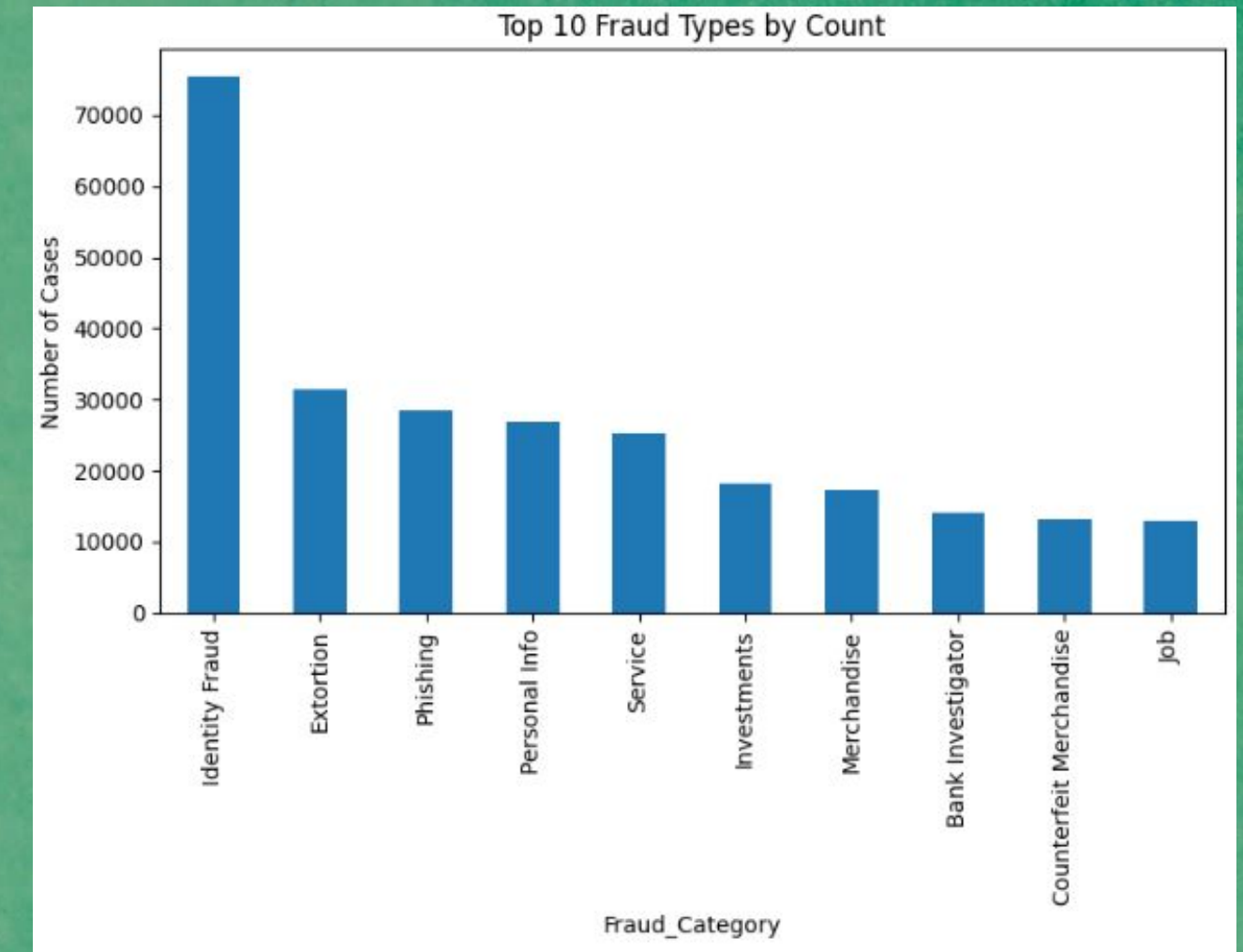
This decline may reflect increased public awareness, better fraud prevention strategies, or changes in reporting behavior.



Top Fraud Categories

Frequent scams aren't always the most damaging: high-volume frauds tend to have lower average losses.

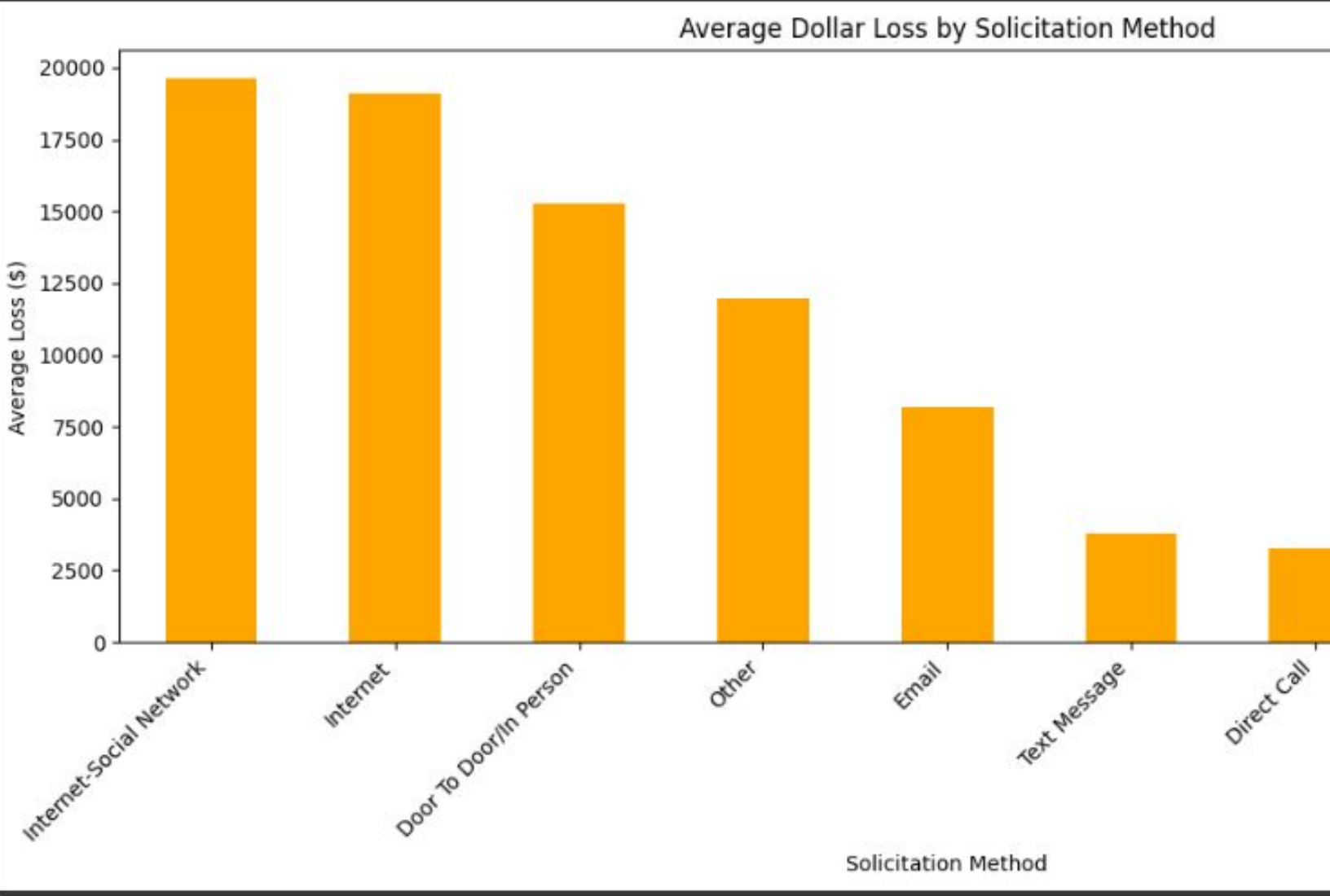
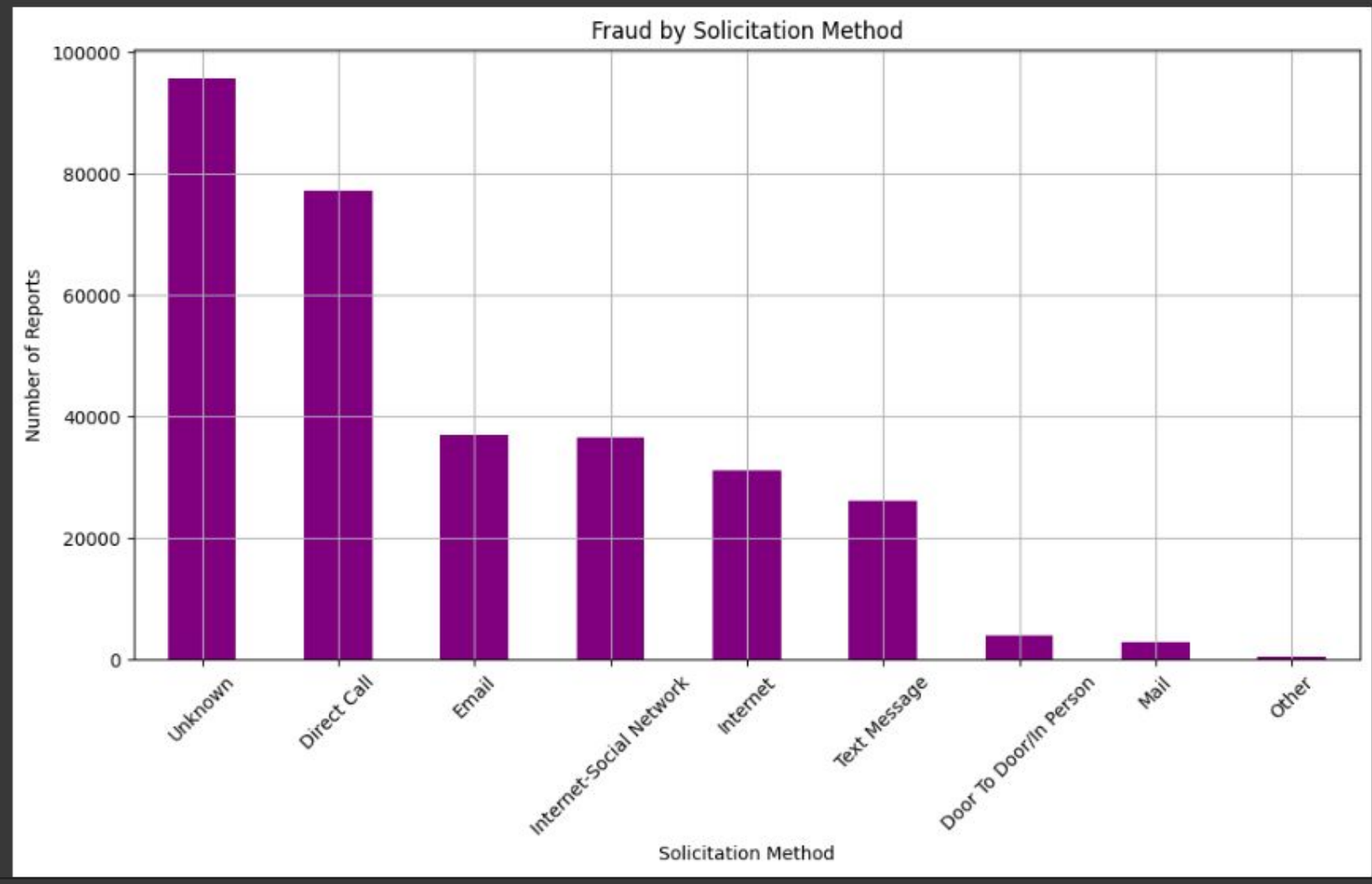
Low-frequency scams like Investment and Romance frauds, although less common, cause severe financial harm per victim. This calls for targeted awareness and prevention efforts.



Solicitation Methods by Frequency

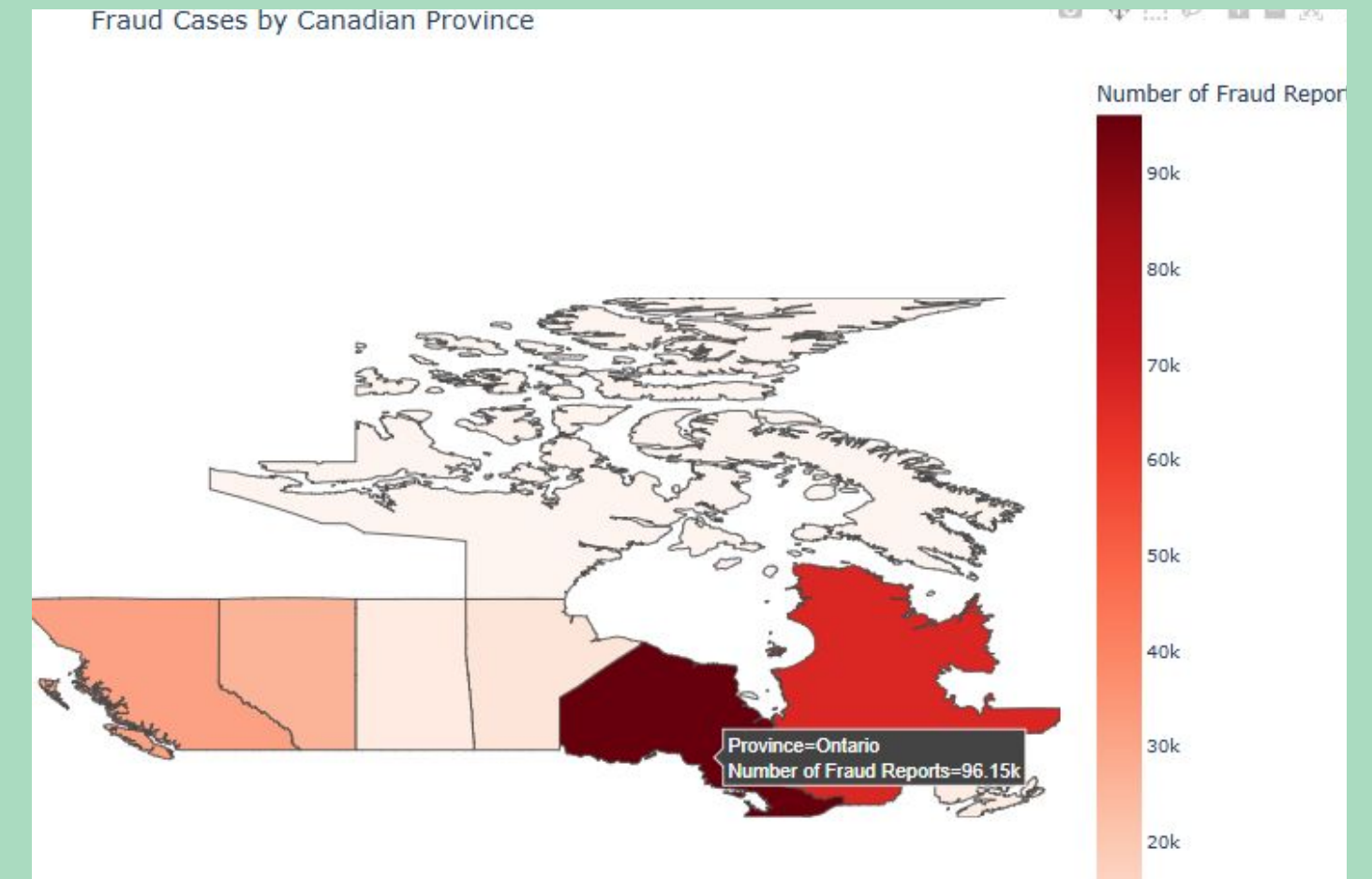
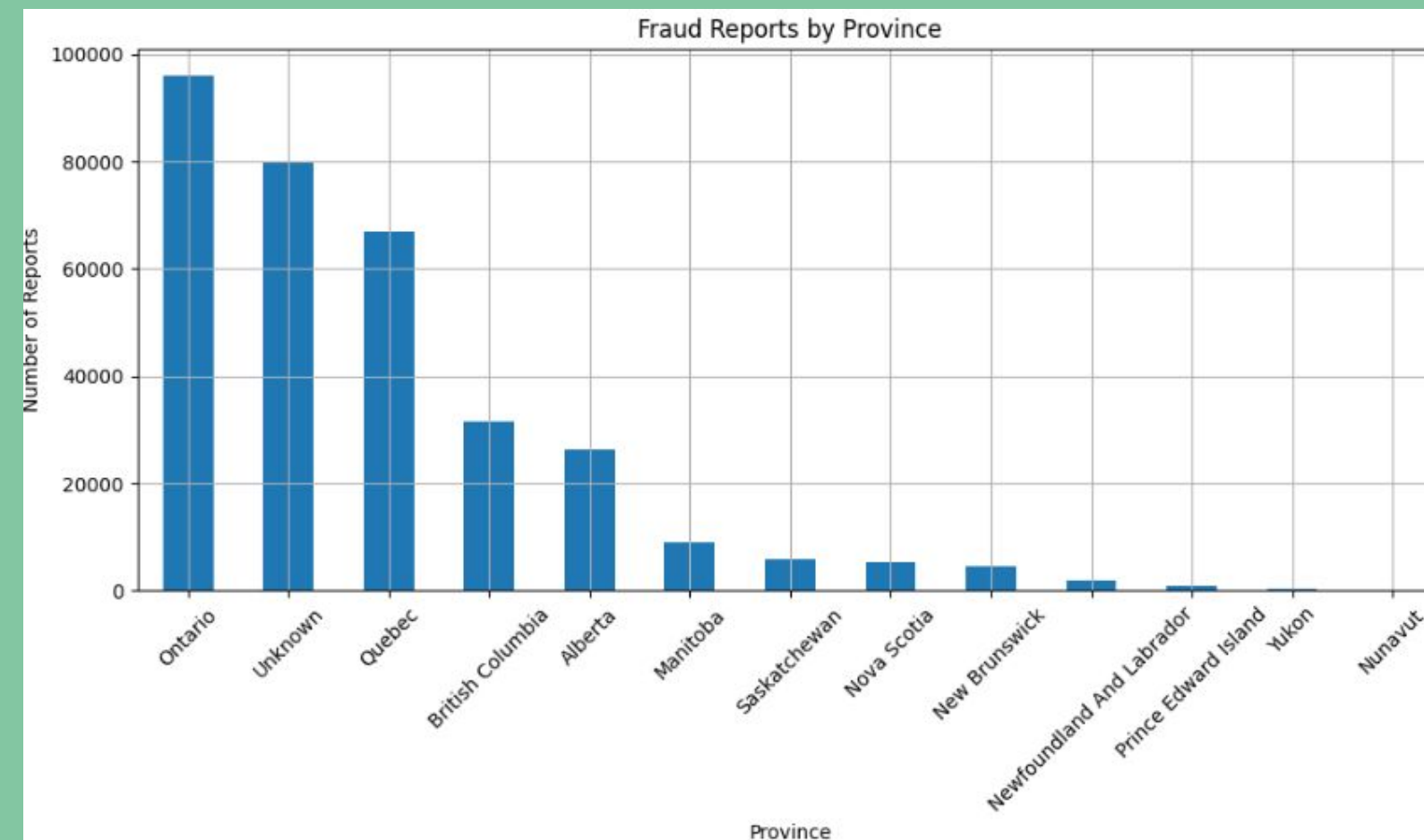
The most frequent solicitation method is Direct Call, followed by Email and Internet-based methods. However, Email and Social Media tend to result in higher average dollar losses, suggesting more successful scams.

A large portion of records is still classified as Other/unknown, indicating room for improving fraud categorization and reporting.



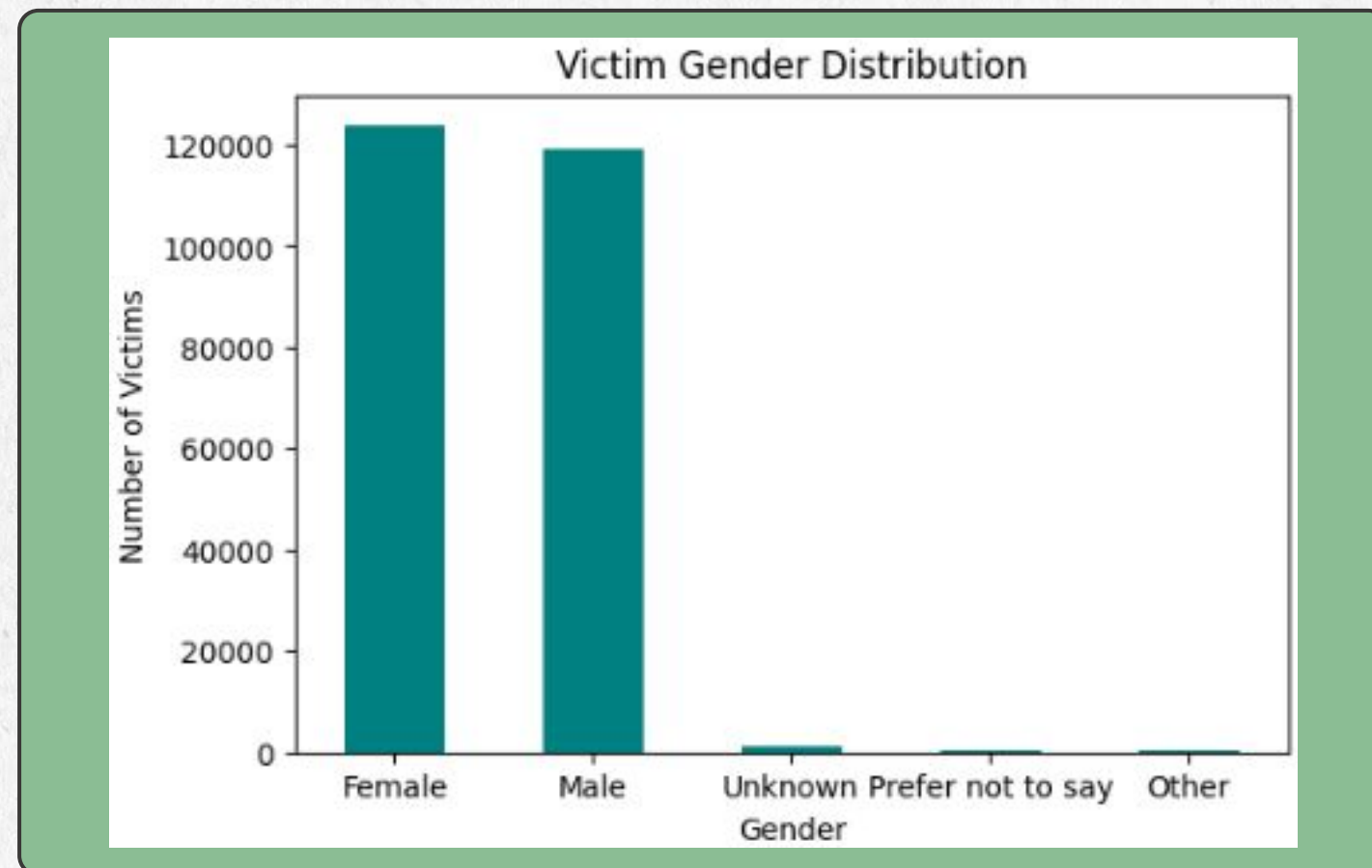
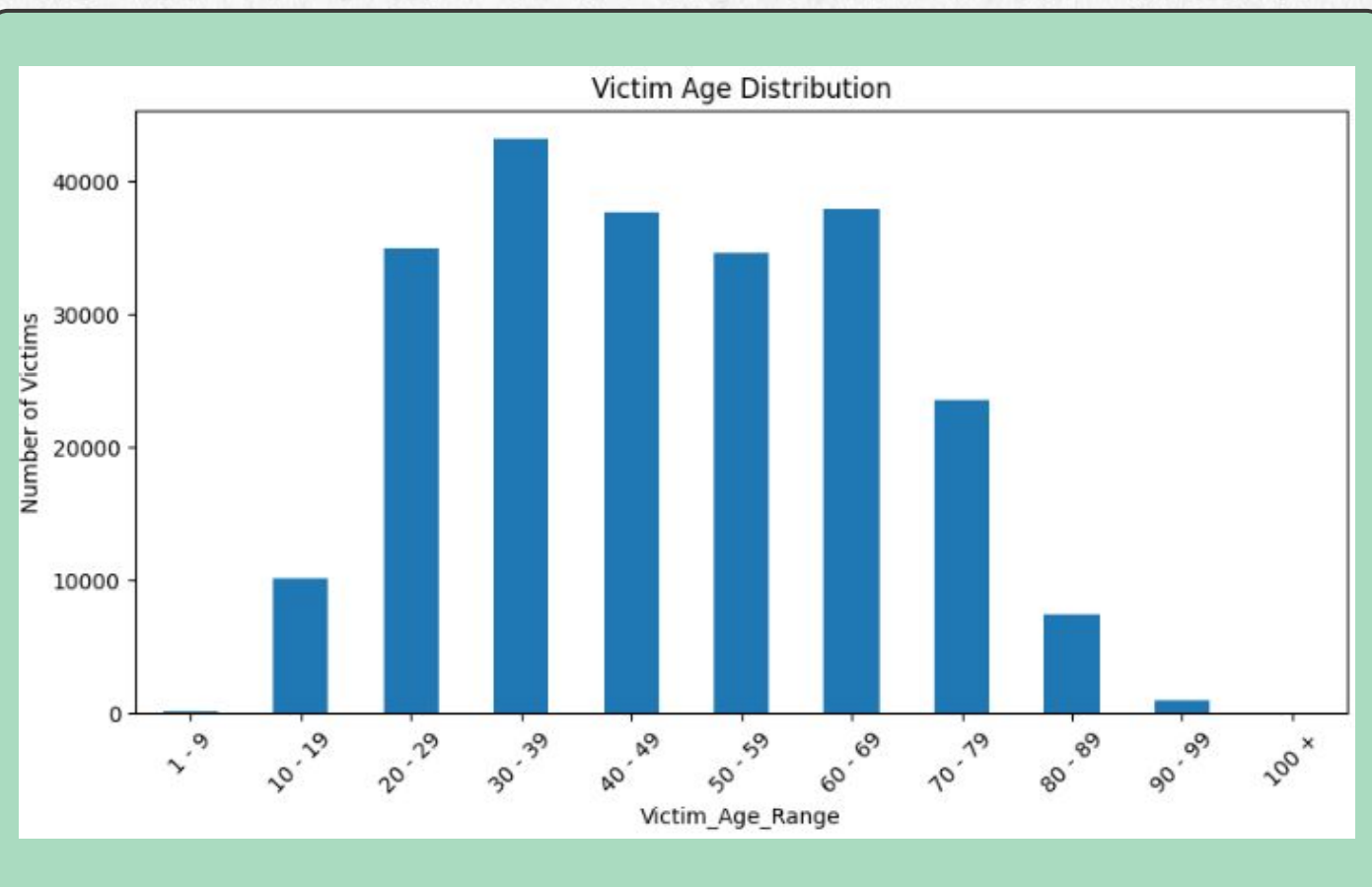
Regional Distribution of Fraud in Canada

Fraud activity appears to be more concentrated in the most populous provinces, likely due to higher population density and more digital/financial activity.



Demographics Distribution

- Adults aged 30–69 are the most frequently targeted group, likely due to higher financial activity and online presence.
- Young adults (20–29) and seniors (60–79) are also major targets.
- Very few cases in children (1–9) and centenarians (100+).
- This suggests a need for fraud prevention messaging tailored to working-age adults and seniors, with extra attention to digital literacy for youth and elderly groups.



Predictive Modeling

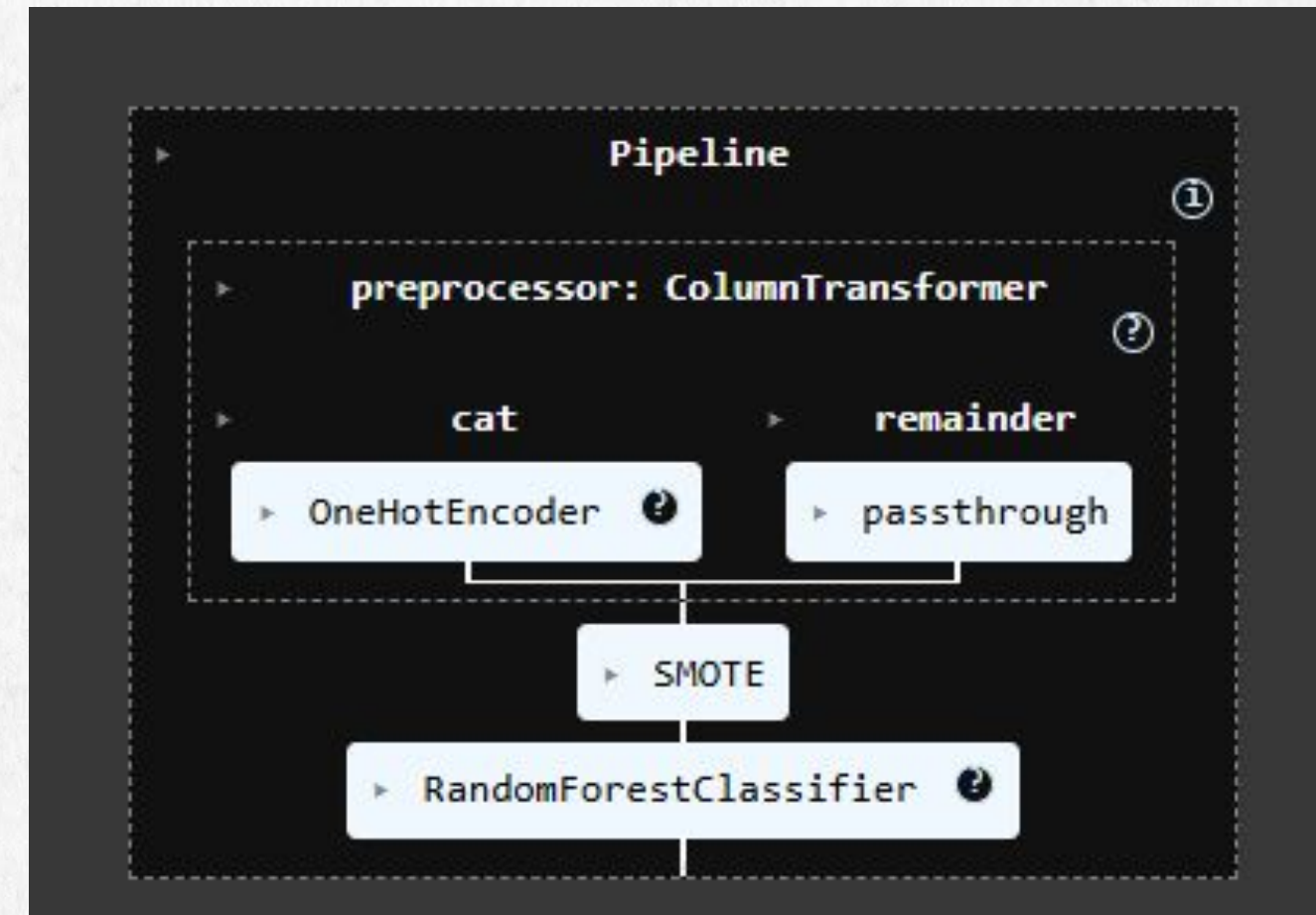
Build models to predict whether a fraud report led to financial loss.

Models Used:

- Logistic Regression
- Decision Tree
- Random Forest

Why these models?

- Interpretable (Logistic, Tree)
- Handles complexity & non-linearity (Random Forest)



Model Performance Comparison

Random Forest: Best balance of recall and precision

Logistic Regression: Best for minimizing false negatives

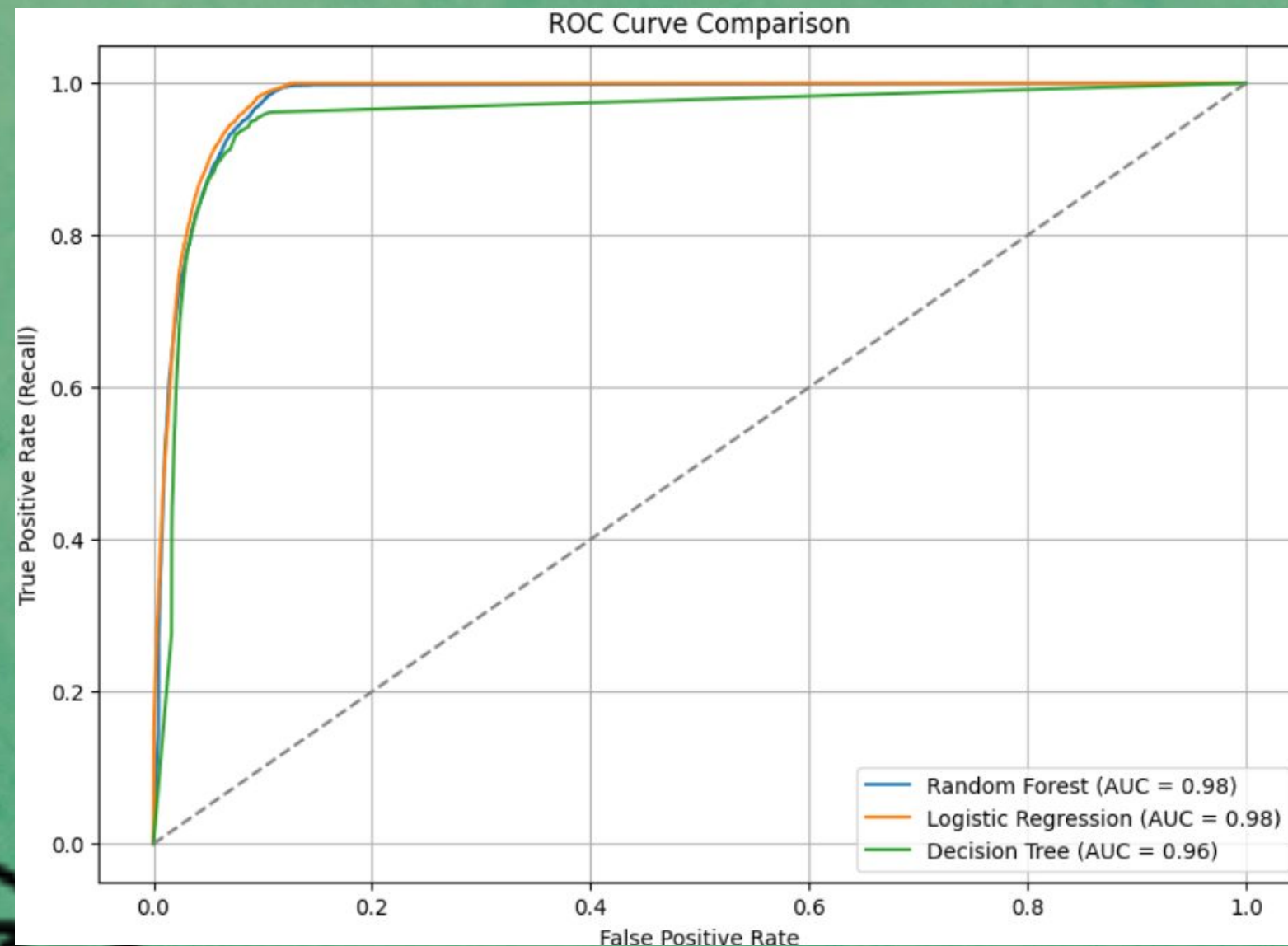
Decision Tree: Most interpretable, but slightly lower performance

Metric / Model	Random Forest	Logistic Regression	Decision Tree
Accuracy	93%	92%	93%
Recall (Loss)	0.94	0.98	0.91
Precision (Loss)	0.81	0.77	0.81
F1-score (Loss)	0.87	0.86	0.86
False Negatives	1018	290	1406
False Positives	3636	4767	3481

Scenario	Best Model	Why
Catch as many fraud losses as possible (low FN)	✓ Logistic Regression	Highest recall (0.98), lowest false negatives (291)
Balanced precision & recall, low FP	✓ Random Forest	Best trade-off between recall (0.94) and precision (0.81)
Simple, interpretable model needed	✓ Decision Tree	Slightly lower performance, but easier to visualize and explain



ROC Curve & AUC Scores

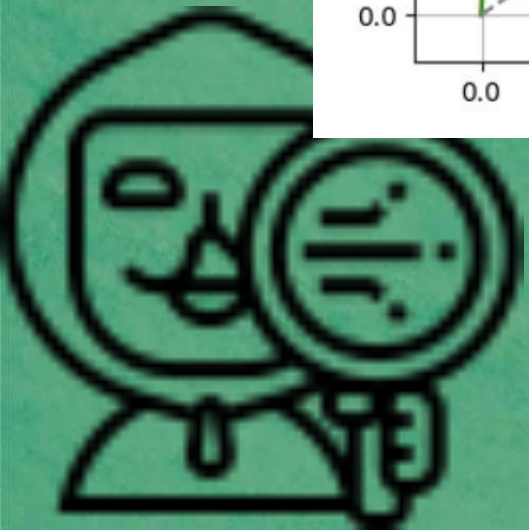


All models perform well, but Random Forest offers best trade-off between sensitivity and specificity

ROC curve shows model's ability to distinguish between fraud with loss vs. no loss

AUC (Area Under Curve) Scores:

- Random Forest: 0.98
- Logistic Regression: 0.98
- Decision Tree: 0.96



Final Conclusion

Trend: Financial fraud has shown significant fluctuations between 2021 and 2025.

Fraud Types: Certain fraud types (e.g., investment scams, phishing) caused the highest losses.

Geographic Impact: Specific provinces experienced disproportionately high fraud cases.

Gender Distribution: Fraud affected both males and females nearly equally, with a slightly higher number of female victims. Very few cases were reported by individuals identifying as other genders or preferring not to say.

Age Demographics: Individuals aged 30–69 were the most targeted by fraud, with the peak in the 30–39 age range. Very young and very elderly age groups experienced significantly fewer incidents.

Random Forest is our recommended model for fraud detection.



Recommendations

- Target education toward adults 30–69 and high-loss fraud types
- Focus campaigns on high-report provinces
- Use predictive modeling (Random Forest) to support fraud prevention efforts
- Explore time-series forecasting and unsupervised anomaly detection in future work



The background is a solid green color with a subtle texture. It is decorated with various white line-art icons related to finance and business. In the top left, there is a stack of papers, a dollar sign, a calendar, and two coins. In the top right, there is a hand holding a stack of papers, a dollar sign, and a coin. In the bottom left, there is a bar chart with three bars of increasing height, a dollar sign, and a coin. In the bottom right, there is a hand holding three coins, a dollar sign, and a calendar. In the center, the text "Thank you!" is written in a large, white, sans-serif font.

Thank you!